

Two categorical variables

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Task 1: First load the packages we may need: tidyverse, ggdist, and ggribes. Install them first if necessary.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.4
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggdist)
library(ggribes)
```

```
##
## Attaching package: 'ggribes'
##
## The following objects are masked from 'package:ggdist':
##
##   scale_point_color_continuous, scale_point_color_discrete,
##   scale_point_colour_continuous, scale_point_colour_discrete,
##   scale_point_fill_continuous, scale_point_fill_discrete,
##   scale_point_size_continuous
```

Today, we will work with the GSS_clean dataset that contains data from the General Social Survey, which is a nationally representative survey of Americans to “monitor and explain trends in opinions, attitudes, and behaviors”.

Task 2: I have put a file called GSS_clean into the data folder in this project. Load the dataset here, and call it GSS:

```
GSS<- read_csv("data/GSS_clean.csv")
```

```
## Rows: 2348 Columns: 45
## -- Column specification -----
## Delimiter: ","
## chr (34): number_of_children, age_of_respondent, college_major_1, college_ma...
## dbl (11): respondent_id_number, highest_year_of_school_completed, highest_ye...
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.  
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Task 3: Run the code below to drop the level NA from the variables we will be using today.

```
GSS2<- GSS %>%  
  filter(marital_status != "NA") %>%  
  filter(general_happiness != "NA") %>%  
  filter(political_party_affiliation != "NA") %>%  
  droplevels()
```

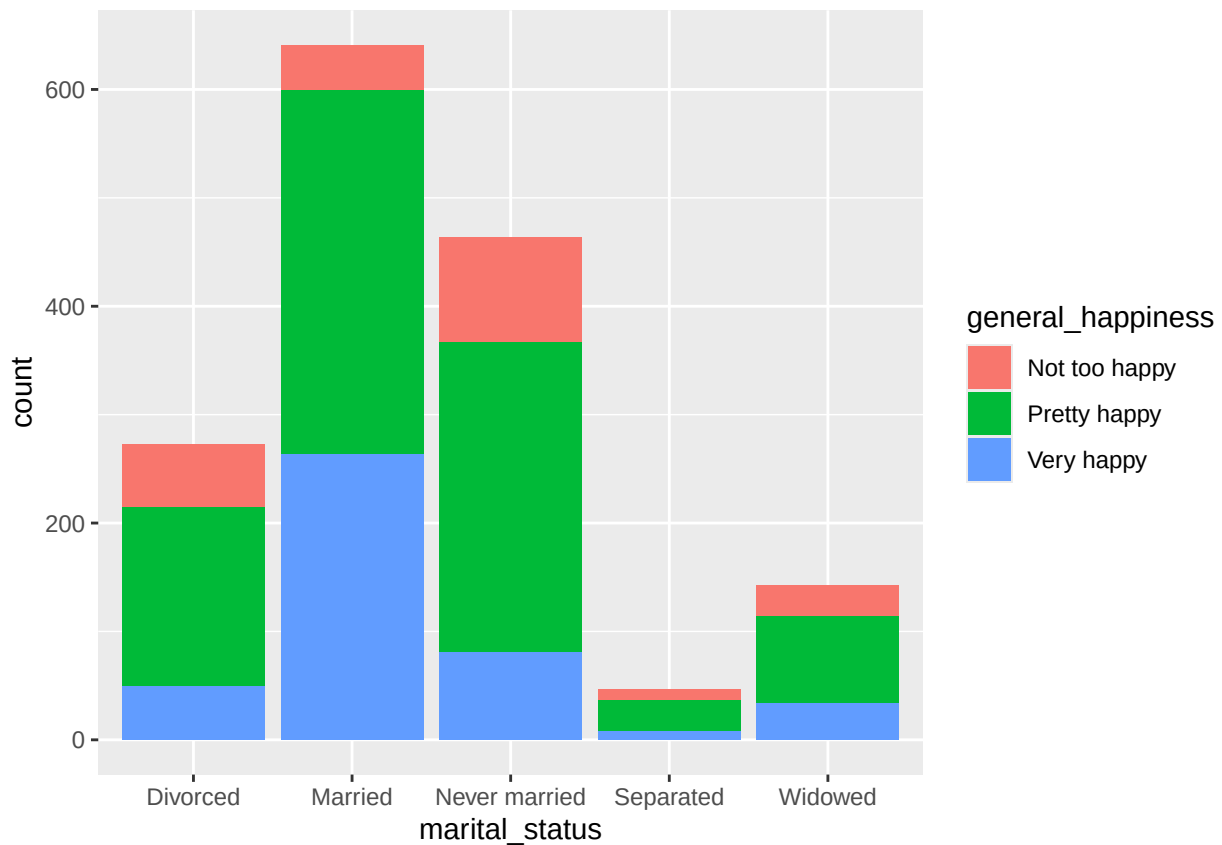
Remember to use GSS2 from now on!

Visualization of two categorical variables

Here, we discuss how to visualize graphs incorporating two categorical variables. Suppose I am interested in exploring the connection between the variables `marital_status` and `general_happiness`. Suppose, in particular, that I am considering `marital_status` as the explanatory variable and `general_happiness` as the response variable. One way to graph this is to use a *stacked bar plot*. To make one of these with vertical bars, then I should put `marital_status` on the x-axis, and fill using the `general_happiness` variable. We will use `geom_bar()`.

Task 4: Fill in the code below to make a stacked bar plot of marital status, filled using `general_happiness`. (Re-read the previous paragraph!)

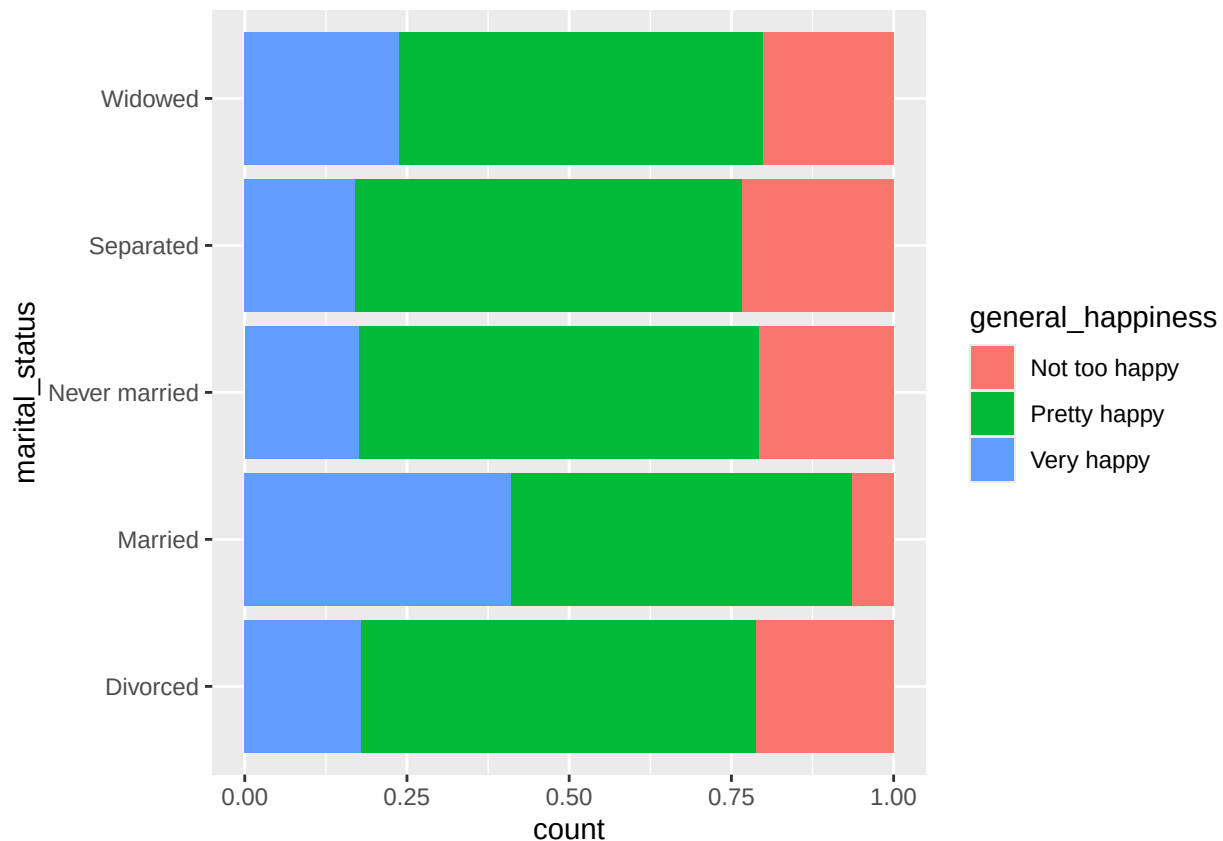
```
ggplot(data=GSS2,  
       mapping = aes(x = marital_status,  
                     fill= general_happiness)) + geom_bar()
```



Notice that we are using `geom_bar` so that R does the counting for us!

Task 5: Suppose I decide I want to put the explanatory variable on the y-axis instead. Then I map `marital_status` to the y-axis, but keep the fill as `general_happiness`. Enter the appropriate code below.

```
ggplot(data=GSS2,
       mapping = aes(y = marital_status,
                     fill = general_happiness)) +
  geom_bar(position = "fill")
```



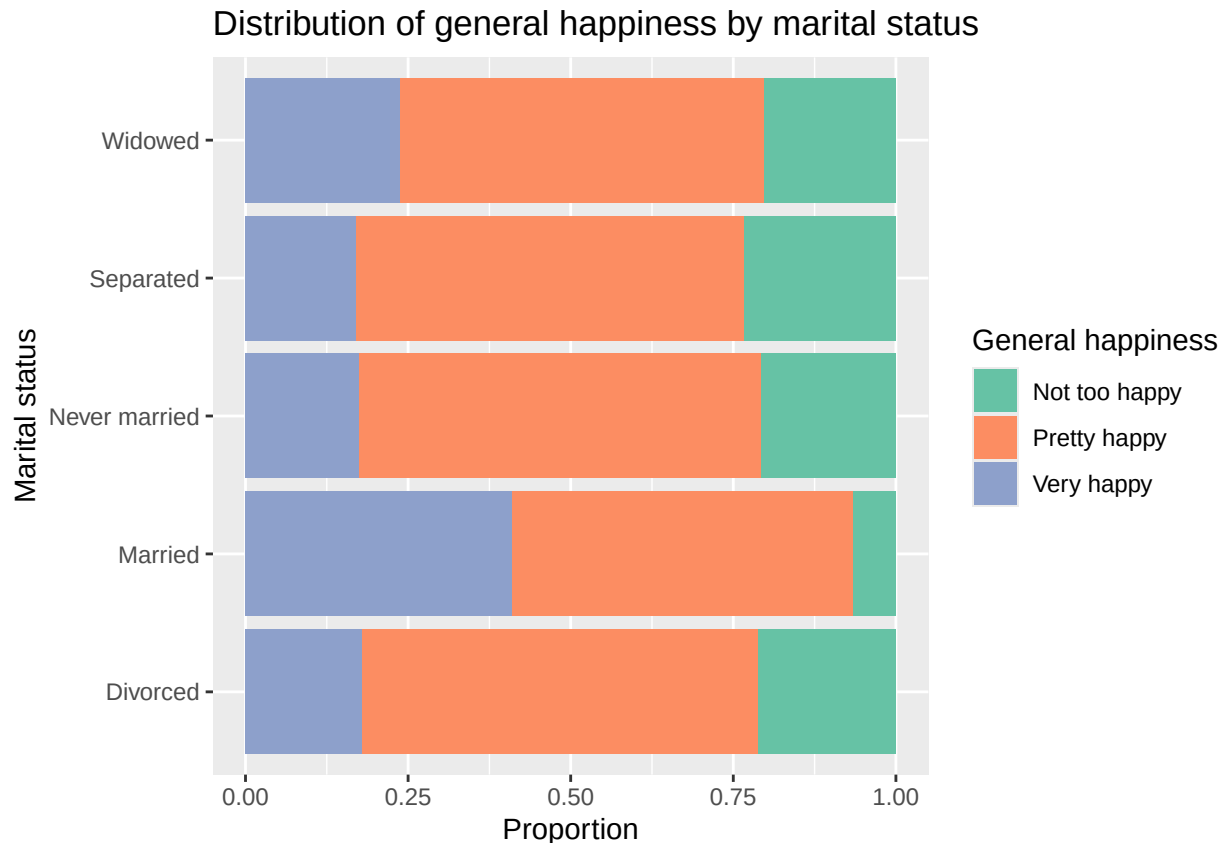
Task 6: Apart from moving `marital_status` to the y-axis, what else is different about this graph compared to the first graph? How did I accomplish this?

Instead of showing raw counts, the y-axis now represents proportions ranging from 0 to 1 making it easier to compare differences in happiness levels across marital statuses. This was done using `position = "fill"` inside `geom_bar()`, which normalizes the bars to show proportions instead of counts.

Often, using proportions in a stacked bar graph makes for a better visualization than just using counts. It is easier to compare the differences between the levels of `marital_status` this way.

Task 7: Below, copy and paste the code from above and make some improvements: add better colors, add axis labels and a title to your graph.

```
ggplot(data = GSS2,
       mapping = aes(y = marital_status,
                     fill = general_happiness)) +
  geom_bar(position = "fill") + scale_fill_brewer(palette = "Set2") +
  labs(
    title = "Distribution of general happiness by marital status",
    x = "Proportion",
    y = "Marital status",
    fill = "General happiness"
  )
```



Task 8: Is there an association between people’s marital status and their level of general happiness, do you think? Explain, supporting what you say using numbers from the graph.

Yes, there appears to be an association between marital status and general happiness based on the graph. For instance, married individuals have the highest proportion of “Very happy” responses, suggesting that marriage is associated with greater happiness compared to other groups. In contrast, divorced, separated, and widowed individuals have a higher proportion of “Not too happy” responses compared to married individuals, indicating that they are generally less happy.

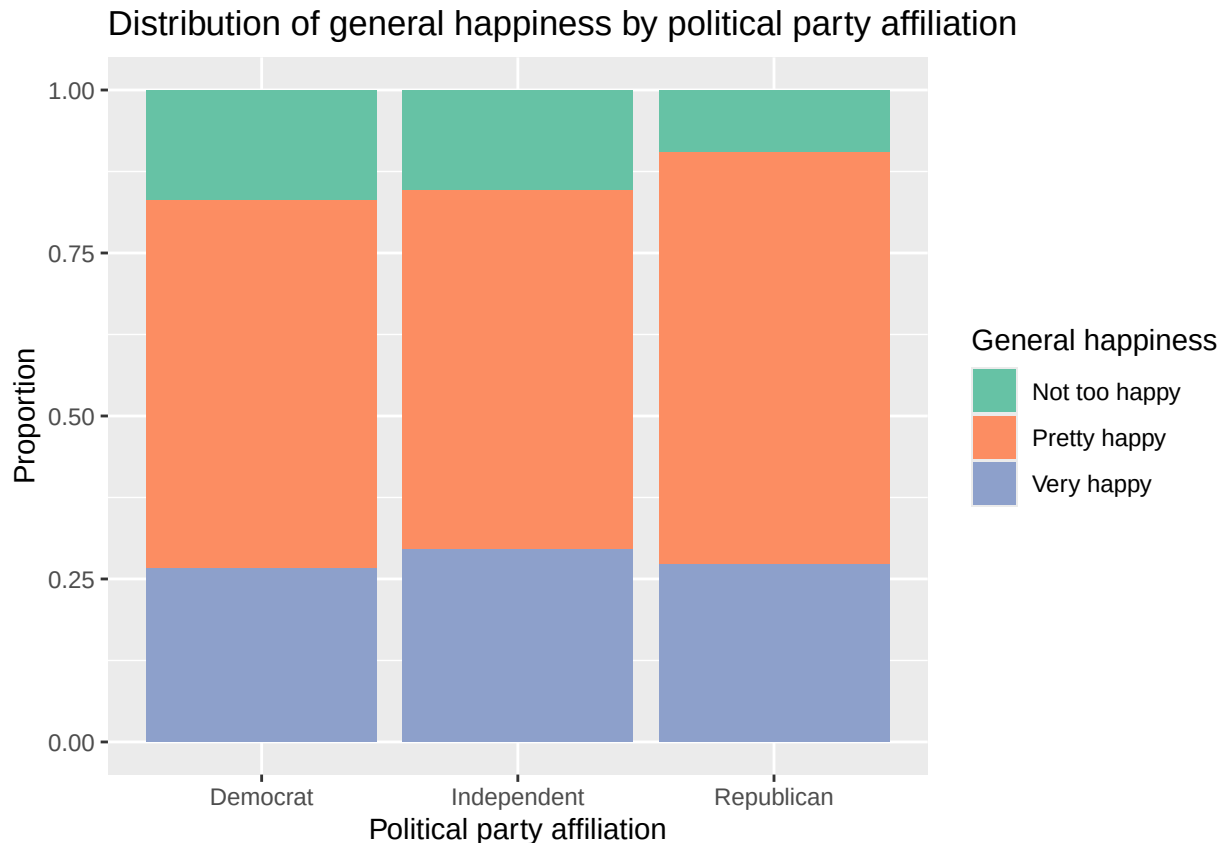
Task 9: I am interested in knowing if people’s political party affiliation has an effect on their level of general happiness. Create an appropriate visualization to check for such an association. (Which variable am I thinking of as explanatory? How will this affect your visualization?)

```
glimpse(GSS2)
```

```
## Rows: 1,567
## Columns: 45
## $ respondent_id_number      <dbl> 1, 2, 3, 4, 6, 7, 8, 10, 12, ~
## $ number_of_children        <chr> "0", "3", "2", "2", "2", "6", ~
## $ age_of_respondent         <chr> "43", "74", "42", "63", "67", ~
## $ highest_year_of_school_completed <dbl> 14, 10, 16, 16, 16, 13, 12, 1~
## $ highest_year_school_completed_father <dbl> 12, 0, 12, 16, 12, NA, NA, 6, ~
## $ highest_year_school_completed_mother <dbl> 12, 8, 16, 12, 12, 6, 12, 20, ~
## $ highest_year_school_completed_spouse <dbl> NA, NA, 14, 14, NA, NA, NA, 1~
## $ college_major_1           <chr> "Communications/speech", NA, ~
## $ college_major_2           <chr> NA, NA, NA, "Computer science~
## $ diploma_ged_or_other      <chr> "High school diploma", "Ged", ~
```

```
## $ respondents_sex <chr> "Male", "Female", "Male", "Fe~
## $ number_of_brothers_and_sisters <dbl> 4, 4, 2, 3, 1, 7, 2, 2, 2, 9,~
## $ labor_force_status <chr> "Temp not working", "Retired"~
## $ number_of_hours_worked_last_week <dbl> NA, NA, 40, 40, NA, 35, 89, 4~
## $ number_of_hours_usually_work_a_week <dbl> 41, NA, NA, NA, NA, NA, NA, N~
## $ self_emp_or_works_for_somebody <chr> "Someone else", "Someone else~
## $ govt_or_private_employee <chr> "Private", "Private", "Privat~
## $ occupation_code <chr> "Human resources workers", "P~
## $ marital_status <chr> "Never married", "Separated",~
## $ marital_type <chr> NA, NA, "Marriage between a m~
## $ race_of_respondent <chr> "White", "White", "White", "W~
## $ born_in_us <chr> "Yes", "Yes", "Yes", "Yes", "~
## $ should_govt_reduce_income_differences <chr> NA, "No govt action", "3", "5~
## $ favor_or_oppose_death_penalty_for_murder <chr> "Favor", "Oppose", "Favor", "~
## $ should_marijuana_be_made_legal <chr> NA, "Not legal", "Legal", "No~
## $ rs_religious_preference <chr> "Christian", "Catholic", "Non~
## $ specific_denomination <chr> "No denomination", NA, NA, "U~
## $ other_protestant_denominations <chr> NA, NA, NA, NA, NA, NA, NA, "~
## $ general_happiness <chr> "Pretty happy", "Very happy",~
## $ condition_of_health <chr> "Good", "Excellent", NA, NA, ~
## $ is_life_exciting_or_dull <chr> "Routine", "Exciting", NA, NA~
## $ ballot_used_for_interview <chr> "Ballot a", "Ballot c", "Ball~
## $ parks_and_recreation <chr> "Too much", "About right", "A~
## $ mass_transportation <chr> "About right", "About right",~
## $ number_of_persons_in_household <dbl> 5, 2, 4, 2, 2, 2, 1, 2, 4, 1,~
## $ total_family_income <chr> "Refused", "$25000 or more", ~
## $ respondents_income <chr> "Refused", "(Missing)", "$250~
## $ region_of_interview <chr> "New England", "New England",~
## $ size_of_place_in_1000s <dbl> 14, 14, 14, 14, 14, 24, 24, 5~
## $ political_party_affiliation <chr> "Republican", "Independent", ~
## $ think_of_self_as_liberal_or_conservative <chr> "Conservative", "(Missing)", ~
## $ space_exploration_program <chr> "Too much", NA, "Too little",~
## $ improving_protecting_environment <chr> "About right", NA, "Too littl~
## $ improving_protecting_nations_health <chr> "Too little", NA, "Too little~
## $ lived_in_us <dbl> NA, NA, NA, NA, NA, NA, NA, N~
```

```
ggplot(data = GSS2,
  mapping = aes(x = political_party_affiliation,
    fill = general_happiness)) +
geom_bar(position = "fill") + scale_fill_brewer(palette = "Set2") +
labs(
  title = "Distribution of general happiness by political party affiliation",
  x = "Political party affiliation",
  y = "Proportion",
  fill = "General happiness" )
```



Task 10: After you have created your graph, write about what you think: is there an association between political party affiliation appear and people's general happiness? Explain, supporting your answer using numbers from the graph.

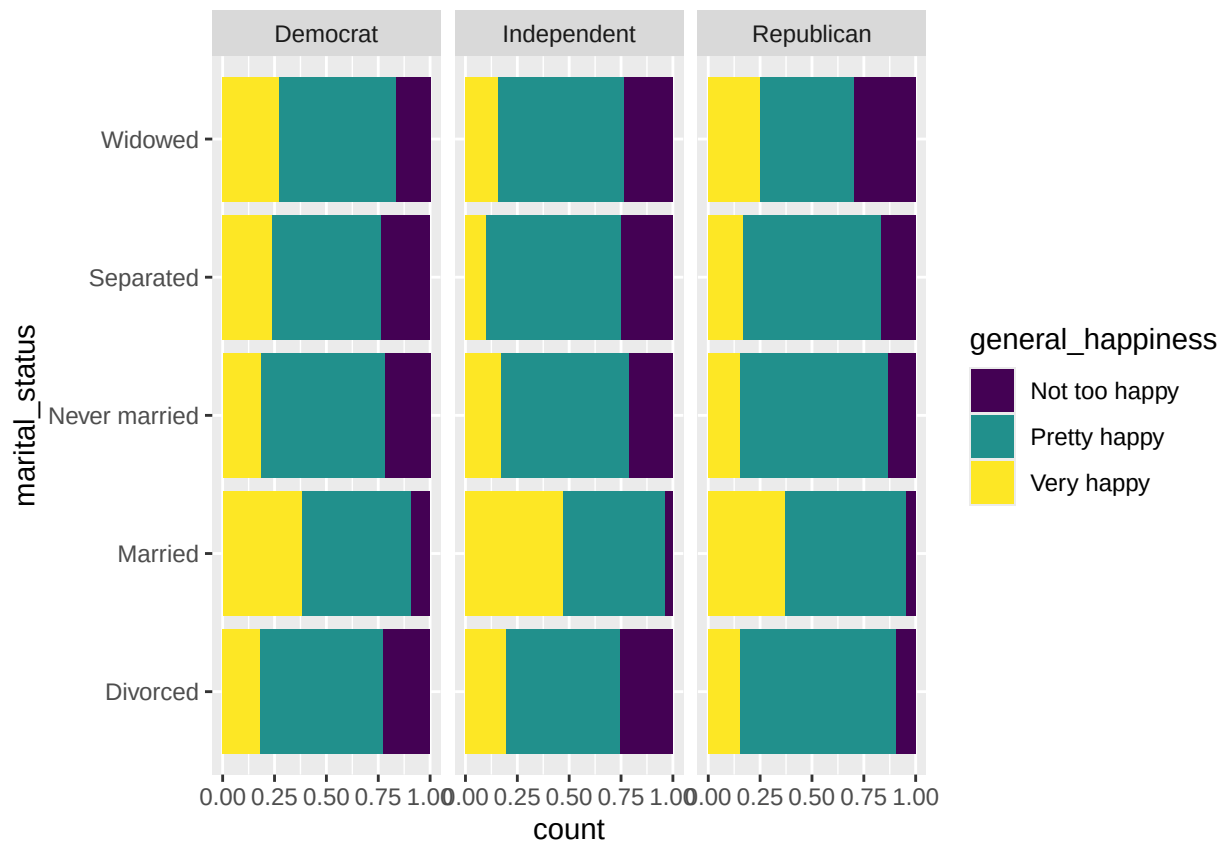
The graph above shows the distribution of responses by political affiliation. While the differences are not drastic, we can observe that republicans tend to report higher levels of happiness, with a greater proportion of them identifying as 'Pretty happy' and 'Very happy' compared to individuals with other political affiliations. For example, roughly 70% of Republicans reported being 'pretty happy,' compared to roughly 60% of Democrats and roughly 55% of independents.

Using faceting to add a third categorical variable

Here is a way to use faceting (making lots of little sub-graphs) to add a THIRD categorical variable to your visualization.

Task 11: Insert the extra line of code `facet_wrap(~political_party_affiliation)` in the specified space below, using `+` symbols as appropriate to add the layers. Run the code to see what happens.

```
ggplot(data=GSS2, mapping = aes(y = marital_status,
                                fill=general_happiness))+
  geom_bar(position = "fill") + facet_wrap(~political_party_affiliation) +
  scale_fill_viridis_d()
```



Contingency Tables

Remember from our first categorical variables lab, here is the code for creating a frequency table of a single variable. Run the code chunk to see what happens.

```
GSS2 %>%
  group_by(marital_status)%>%
  summarize(n = n())
```

```
## # A tibble: 5 x 2
##   marital_status      n
##   <chr>          <int>
## 1 Divorced        273
## 2 Married         641
## 3 Never married   463
## 4 Separated        47
## 5 Widowed        143
```

We can add to this to create a table containing information about two categorical variables. Run the code.

```
GSS2 %>%
  group_by(marital_status, general_happiness)%>%
  summarize(n = n())
```

```
## `summarise()` has grouped output by 'marital_status'. You can override using
## the ``.groups` argument.
```

```
## # A tibble: 15 x 3
## # Groups:   marital_status [5]
```



```
## marital_status general_happiness n
## <chr> <chr> <int>
## 1 Divorced Not too happy 58
## 2 Divorced Pretty happy 166
## 3 Divorced Very happy 49
## 4 Married Not too happy 42
## 5 Married Pretty happy 336
## 6 Married Very happy 263
## 7 Never married Not too happy 96
## 8 Never married Pretty happy 286
## 9 Never married Very happy 81
## 10 Separated Not too happy 11
## 11 Separated Pretty happy 28
## 12 Separated Very happy 8
## 13 Widowed Not too happy 29
## 14 Widowed Pretty happy 80
## 15 Widowed Very happy 34
```

Task 12: Judging from the table you made in the previous code chunk, how many people in the dataset, in total, are divorced? How many are married? (Show how you get your answers!)

There are 273 divorced individuals and 641 married individuals in the dataset. I determined this by summing the frequency for each divorced and married response in the table above. As a note, there is a way to sum values based on the category name. However, since the table was relatively small, I computed it by hand.

This kind of table is a step in the right direction, but not ideal. Next, we will work towards making an improved version of this table.

Next, we add another column containing proportions. Run the code below.

```
GSS2 %>%
  group_by(marital_status, general_happiness)%>%
  summarize(n = n()) %>%
  mutate(prop = n/sum(n))

## `summarise()` has grouped output by 'marital_status'. You can override using
## the `.groups` argument.

## # A tibble: 15 x 4
## # Groups:   marital_status [5]
## marital_status general_happiness n prop
## <chr> <chr> <int> <dbl>
## 1 Divorced Not too happy 58 0.212
## 2 Divorced Pretty happy 166 0.608
## 3 Divorced Very happy 49 0.179
## 4 Married Not too happy 42 0.0655
## 5 Married Pretty happy 336 0.524
## 6 Married Very happy 263 0.410
## 7 Never married Not too happy 96 0.207
## 8 Never married Pretty happy 286 0.618
## 9 Never married Very happy 81 0.175
## 10 Separated Not too happy 11 0.234
## 11 Separated Pretty happy 28 0.596
## 12 Separated Very happy 8 0.170
## 13 Widowed Not too happy 29 0.203
## 14 Widowed Pretty happy 80 0.559
```

```
## 15 Widowed          Very happy          34 0.238
```

Task 13: Examine the proportions that R has calculated. Can you figure out their meaning? For example, can you write a sentence to explain the meaning of the proportion 0.19834711 in the first row? What is it a proportion of?

The proportion 0.19834711 is not in the dataset (not sure if I did it right, plan on correcting Friday after exam).

The proportion in the first row represents the fraction of all individuals within the ‘Separated’ group who reported being ‘Pretty happy’. This means that approximately 59% of Separated individuals in the dataset identified as ‘Pretty happy’.

Below is code to create a much nicer-looking table. Pay attention to what parts of the code are the same as above, and what is new.

```
GSS2 %>%
  group_by(marital_status, general_happiness)%>%
  summarize(n=n()) %>%
  mutate(prop = n/sum(n))%>%
  pivot_wider(id_cols = general_happiness,
              names_from = marital_status,
              values_from = n)
```

```
## `summarize()` has grouped output by 'marital_status'. You can override using
## the `groups` argument.
```

```
## # A tibble: 3 x 6
##   general_happiness Divorced Married `Never married` Separated Widowed
##   <chr>             <int>  <int>         <int>      <int>  <int>
## 1 Not too happy      58    42           96         11     29
## 2 Pretty happy     166   336          286         28     80
## 3 Very happy        49   263           81          8     34
```

Task 14: What is different about this table compared to the previous one? How did we accomplish this?

This new table reshapes the data so that `general_happiness` remains as row labels, while `marital_status` is converted into multiple columns. This makes it easier to compare happiness levels across marital statuses, rather than having a long list of counts. This was accomplished by setting the following: “`names_from = marital_status`” in the `pivot_wider` function.

Task 15: Next, have R put the column *proportions* in the table, instead of the counts. To do this, put `prop` instead of `n` in the `values_from` argument.

```
GSS2 %>%
  group_by(marital_status, general_happiness)%>%
  summarize(n=n()) %>%
  mutate(prop = n/sum(n))%>%
  pivot_wider(id_cols = general_happiness,
              names_from = marital_status,
              values_from = prop)
```

```
## `summarize()` has grouped output by 'marital_status'. You can override using
## the `groups` argument.
```

```
## # A tibble: 3 x 6
```

```
##   general_happiness Divorced Married `Never married` Separated Widowed
##   <chr>             <dbl>   <dbl>             <dbl>   <dbl>   <dbl>
## 1 Not too happy    0.212  0.0655             0.207   0.234   0.203
## 2 Pretty happy     0.608  0.524              0.618   0.596   0.559
## 3 Very happy       0.179  0.410              0.175   0.170   0.238
```

Notice, in this table, the variable `marital_status` labels the columns. It is usual for the *explanatory variable* to label the *columns* in a table like this and you should do the same when you are creating tables. The proportions displayed in the table above are “column proportions”.

Task 16: How am I able to tell, by just looking at the table that these are column proportions?

We can tell these are column proportions because each column sums to exactly one.

Task 17: Explain the meaning of the proportion 0.6122754 in the table.

The proportion 0.6122754 means that 61.23% of individuals within the ‘never married’ category reported being pretty happy. This tells us that a majority of people who have never been married fall into this happiness category.

Task 18: Create a new contingency table of proportions for `political_party_affiliation` and `general_happiness`, thinking of `political_party_affiliation` as the explanatory variable.

```
GSS2 %>%
  group_by(political_party_affiliation, general_happiness) %>%
  summarize(n = n(), .groups = "drop") %>%
  mutate(prop = n / sum(n), .by = political_party_affiliation) %>%
  pivot_wider(id_cols = general_happiness,
              names_from = political_party_affiliation,
              values_from = prop)
```

```
## # A tibble: 3 x 4
##   general_happiness Democrat Independent Republican
##   <chr>             <dbl>     <dbl>         <dbl>
## 1 Not too happy    0.169     0.154     0.0956
## 2 Pretty happy     0.565     0.551     0.632
## 3 Very happy       0.266     0.295     0.272
```